1.7 Linear Independence

Definition: An indexed set of vectors $\{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ is said to be linearly independent if the vector equation

$$x_1v_1 + x_2v_2 + \cdots + x_pv_p = 0 \qquad (1)$$

has only the trivial solution ($x=0$). The set $\{v_1, \dots, v_p\}$ is said to be linearly dependent if there exist weights $c_1, \dots, c_p$, not all zero, such that

$$c_1v_1 + c_2v_2 + \cdots + c_pv_p = 0 \qquad (2)$$

Equation (2) is called a linear dependence relation among $v_1, \dots, v_p$ when weights are not all zero.

For brevity, we may say that $v_1, \dots, v_p$ are linearly dependent when we mean that $\{v_1, \dots, v_p\}$ is a linearly dependent set.

Ex: Let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, $v_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

a. Determine the set $\{v_1, v_2, v_3\}$ is linearly independent.

b. If possible find a linear dependence (relation) among v_1, v_2 and v_3.

a. We must determine if there is a nontrivial solution of the following equation:

$$x_1\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + x_3\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad (3)$$

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Row operations on the associated augmented matrix show that

$$\begin{bmatrix} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Clearly x_1 and x_2 are basic variables and x_3 is free.

Each nonzero value of x_3 determines a nontrivial solution of the given equation (Eq. 3). Hence, v_1, v_2 and v_3 are linearly dependent.

5. To find a linear dependence relation among v_1, v_2 and v_3 completely row reduce the augmented matrix:

$$\begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{aligned} x_1 - 2x_3 &= 0 \\ x_2 + x_3 &= 0 \\ 0 &= 0 \end{aligned} \rightarrow \begin{aligned} x_1 &= 2x_3 \\ x_2 &= -x_3 \\ x_3 &\text{ is free} \end{aligned}$$

To find a linear dependence relation, now, choose any value for x_3 - say $x_3 = 5$. Then $x_1 = 10$, $x_2 = -5$, $x_3 = 5$. Substitute these values into the given equation (Eq. 3) and obtain

$$10v_1 - 5v_2 + 5v_3 = 0$$

$$10v_1 + 5v_3 = 5v_2$$

$$2v_1 + v_3 = v_2$$

This is one (out of infinitely many) possible linear dependence relations among v_1, v_2 and v_3.

Linear Dependence of Matrix Columns

The columns of a matrix A are linearly independent if and only if the equation $Ax = 0$ has only the trivial solution.

Ex:

↳ Determine if the columns of the matrix $A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 5 & 8 & 0 \end{bmatrix}$ are linearly independent.

To study $Ax = 0$, row reduce the augmented matrix:

$$\begin{bmatrix} 0 & 1 & 4 & 0 \\ 1 & 2 & -1 & 0 \\ 5 & 8 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & -2 & 5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 13 & 0 \end{bmatrix}$$

It is clear that there are no free variables. So, $Ax = 0$ has only the trivial solution. Therefore, the columns of A are linearly independent.

Ex: Determine if the following set of vectors are linearly independent.

a. $v_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ $v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

To do for next week.

b. $v_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ $v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

Theorem: An indexed set $S = \{v_1, \dots, v_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others.

Ex:

Theorem: If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set $\{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

Ex:

Theorem: If a set $S = \{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ contains the zero vector, then the set is linearly dependent.